Shahid Hussain

+86 131 8615 1097 | shahidzee5@gmail.com | GitHub | LinkedIn | ORCID | Google Scholar

PROFESSIONAL SUMMARY

AI Researcher and Engineer with 4+ years of experience in computer vision, deep learning, machine learning, and federated learning. I design, develop, and evaluate advanced AI models for medical image analysis, image classification, object detection, segmentation, image enhancement, and denoising using Python, PyTorch, TensorFlow, and OpenCV. I leverage CNNs, Vision Transformers (ViT), Mamba, GANs, and U-Net architectures to build intelligent vision systems, and I investigate Explainable AI (XAI) and Federated Learning to develop trustworthy, interpretable, and privacy-preserving AI solutions. I aim to apply these methodologies to interdisciplinary research across healthcare, industrial automation, robotics, intelligent transportation, and other real-world domains.

EDUCATION

Master of Engineering in Electronics Science & Technology (AI) 2024 – Jun 2026
Xi'an Jiaotong University Xi'an, China

- **Final Grade**: 3.59/4.0
- **Thesis**: Attention-Enhanced CNN-ViT-Mamba Tri-Hybrid for Lung Histopathology Classification
- **Major Courses**: Machine Vision, AI in Medical Imaging, Deep Learning, Image Processing, Signal Processing, Cyber Security

Bachelor of Science in Software Engineering 2018 – 2022
Islamia College University Peshawar, Pakistan

- **Final Grade**: 3.39/4.0
- **Thesis**: XNET: CNN Model for COVID-19 Classification Using Skip Connection
- **Major Courses**: Object-Oriented Programming (OOP), Data Structures and Algorithms, Software Architecture, Artificial Intelligence

PROFESSIONAL EXPERIENCE

Graduate Researcher Computer Vision 2024 – Jun 2026
Xi'an Jiaotong University Xi'an, China

- Build hybrid CNNTransformerMamba pipelines for histopathology image classification; run experiments in PyTorch/TensorFlow with clear metric reporting.
- Lead data work (tiling, annotation, augmentation, splits) and write documentation suitable for publication.
- Parallel projects include face-based attendance (detection recognition pipeline) and early fire detection.

Research Assistant Computer Vision Mar. 2023 – Aug 2024
Digital Image Processing (DIP) Lab Islamia College University Peshawar, Pakistan

- Built XNET: CNN model for Xray image classification; ran experiments in TensorFlow with clear metric reporting.
- Led data work (tiling, annotation, augmentation, splits) and wrote documentation suitable for publication.

Internship Computer Vision Sep. 2022 – Mar. 2023
Bright Code Lab Peshawar, Pakistan

- Data Engineering: Preprocessing, dataset curation, annotation, augmentation.
- Model Optimization: PyTorch, TensorFlow, tuning, ablation, benchmarking.
- Production Integration: Deployment, compression, optimization, and engineering standards.
- Research Communication: Documentation, reporting, presentations, R&D alignment.

PUBLICATIONS

- Submitted Papers
  - **Shahid Hussain**[a], Yi Wenhui[a,*], Hou Jin[b]. "TriH-ACM: A Novel CNNViTMamba Hybrid with Attention for State-of-the-Art Lung Cancer Classification." Submitted to *Biomedical Signal Processing and Control*. **(Revision Submitted)**.
  - **Shahid Hussain**[a], Yi Wenhui[a,*], Tong Jianxing[a], Bagh Hussain[b], Hou Jin[c]. "Advancing Lung Cancer Diagnosis with Hybrid Models: A Histopathology-Based CNN-Transformer Framework." Submitted to *Computer Methods and Programs in Biomedicine*. **(Submitted)**.
  - **Shahid Hussain**[a], Yi Wenhui[a,*], Bagh Hussain[b], Ehtasham Hussain[c], Hou Jin[d]. "A Data-Efficient Vision Transformer with Integrated Explainability for High-Accuracy Lung Cancer Histopathology Classification." Submitted to *Multimedia Tools and Applications*. **(Revision Submitted)**.

- **Shahid Hussain**[a], Yi Wenhui[a,*], Hou Jin[b], Inzimam Hussain[c], Mehak Shafique[c]. "Interpretable Deep Learning for Lung Cancer Diagnostics: A ConvNeXt-GradCam Framework Leveraging Transfer Learning on Histopathological Images." Submitted to *The Journal of Supercomputing*. **(Submitted)**.
- Yi Wenhuia[a,*], **Shahid Hussain**[a], Ehtasham Hussain[b], Inzimam Hussain[c], Mehak Shafique[c], Hou Jin[d]. "Tri-Hybrid State-Space Fusion Networks for Enhanced Multimodal Medical Image Analysis." Submitted to *Pattern Recognition*.. (**Under Review**).
- Jianxing Tong[a], Wenhui Yi[a,*], **Shahid Hussain**[a]. "HCTNet: A Scale-Driven Hybrid CNN-Transformer for Dermoscopic Skin Lesion Classification." **(Submitted)**.

- Published Papers
  - **Shahid Hussain***, Nazeem Khan, Syeda Amna Ali Shah, Shareefa Bano, Abdul Wadood Khan. "AI-Driven Personalized Meal Planning: A Web-Based Platform for Tailored Nutrition and Health Management." *International Journal of Computer Applications*, 187(32):1–8, 2025. DOI: 10.5120/ijca2025925913.
  - **Shahid Hussain***, Hashmat Ullah, Shah Zeb, Abbas Khan. "An Integrated Computer Vision Pipeline for Face-based Attendance: Detection, Recognition, and Secure Logging." *International Journal of Computer Applications*, 187(42):62–68, 2025. DOI: 10.5120/ijca2025925742.
  - **Shahid Hussain**[1,*], Inam Ullah[2], Abdul Wadood Khan[3], Abbas Khan[4], Hamza Khan[5]. "Design and Evaluation of a Convolutional Neural Network for Early Fire Detection in Static Images." *Cognizance Journal of Multidisciplinary Studies*, 5(8):82–95, 2025. DOI: 10.47760/cognizance.2025.v05i08.008.
  - **Shahid Hussain***, Hamza Khan, Ehtasham Hussain, Jamil Ahmad. "XNET: CNN Model for COVID-19 Classification using Skip Connection." *Cognizance Journal of Multidisciplinary Studies*, 4(1):125–134, 2024. DOI: 10.47760/cognizance.2024.v04i01.014.

TECHNICAL SKILLS

- **Programming Languages**: Python, C++, Java, MATLAB
- **Research & Methodologies**: Experimental Design, Literature Review, Data Analysis, Scientific Writing, Paper Publishing, Reproducible Workflows
- **AI & Computer Vision**: Machine Learning, Deep Learning, Hybrid Deep Learning, Vision Transformers (ViT), Mamba (statespace models), YOLOv5, Federated Learning, GANs, Model Optimization, Quantization, Edge AI
- **Python & ML Stack**: Python, TensorFlow, PyTorch, Keras, scikitlearn, Hugging Face, LlamaIndex, OpenCV, NumPy, Pandas, Matplotlib
- **Data & Experimentation**: Data Annotation, Data Augmentation, Dataset Curation, Experiment Design, Model Evaluation & Reporting
- **Tools & Platforms**: Git, Linux, VS Code, PyCharm, LaTeX, Jupyter Notebook, MS Office
- **Interpersonal & Professional**: CrossCultural Collaboration, Teamwork, Adaptability, Excellent Communication, Problem Solving, Task Prioritization

CONFERENCES & SEMINARS

- Belt and Road Key and Core Technology Achievement Trade Fair & Shaanxi Innovation, Entrepreneurship, and Creativity Conference, Xian, China — Nov 2025
- International Training Workshop on Big Data and AI, Xian Jiaotong University, China — Nov 2024
- Belt and Road Key and Core Technology Achievement Trade Fair, Xian, China — Nov 2024

HONORS & AWARDS

- Belt and Road Scholarship Master's Degree (Xian Jiaotong University, China) — Sep 2024 – Jun 2026
- Certificate of Research Experience & Project Completion (Xian Jiaotong University, China) — Feb 2025
- Introduction to Computer Vision and Image Processing (IBM Skills Network) — Mar 2023
- Foundations of Digital Marketing & E-commerce (Google) — Mar 2023
- Programming for Everybody (Python) (University of Michigan, Coursera) — Feb 2023
- Introduction to Artificial Intelligence (AI) (IBM Skills Network) — Dec 2022
- Introduction to TensorFlow for AI, Machine Learning, and Deep Learning (DeepLearning.AI, Coursera) — Dec 2022
- Supervised Machine Learning: Regression and Classification (DeepLearning.AI, Coursera) — Dec 2022
- Data Visualization using Python (Great Learning) — Sep 2022
- Introduction to Machine Learning (Kaggle) — Aug 2022
- Machine Learning Algorithms (Great Learning Academy) — Aug 2022
- The Fundamentals of Digital Marketing (Google Digital Garage) — Oct 2020

LANGUAGES

Chinese Intermediate (HSK 3) | **English** Fluent | **Urdu** Native | **Pashto** Native

REFERENCES

Prof. Yi Wenhui *(Master Supervisor)*
Professor
Xi'an Jiaotong University, China
yiwenhui@mail.xjtu.edu.cn

Prof. Li Guobing *(Course Instructor)*
Associate Professor
Xi'an Jiaotong University, China
gbli@mail.xjtu.edu.cn